

Supplementary Material: Metadata Exclusion in Retrieval Services: Detection and Repair Evidence

References are numbered as in the main article. These supporting measurements use the same artifact and do not constitute independent deployments.

APPENDIX A

AUDIT ROLES AND TRUST BOUNDARY

Figure 5 and Table III detail the trusted-host model. Acquisition and startup must bind histories, writes and the actual reader to the declared state. Receipts identify destinations; hashes identify supplied bytes. Neither authenticates production against a host answering through a compliant copy. Outsider access supplies no assumed corpus, configuration or trace control.

APPENDIX B

SUPPORTING TARGET-SCORE ATTRIBUTION STUDY

Hayhooks 1.24.0 and Haystack 2.31.0 run fetching, conversion, embedding, writing and exact cosine retrieval. Both field policies parse, store and return PatientComments; only embedding inclusion differs. Each 608-document world contains 440 paragraphs from RFC 9110, 72 topical backgrounds and 96 probes. Twelve themes give 36 queries and four pairs/theme (Figure 6). Two field selections, two orders and two phases yield eight fresh processes with paired identities, bodies, membership, queries and order fixed. Normalized float32 MiniLM uses untruncated inputs and revision 1110a243fdf4706b3f48f1d95db1a4f5529b4d41.

Clients use receipts, IDs, scores and order; private vectors reconstruct scores only after decisions. Rank assigns absence $k + 1$ and abstains for two absences. Strict-null gating requires an unchanged control with no detectable effect. The eight worlds produce 152 ingestion requests, 576 searches and 4,864 writes; reconstruction matches all 17,280 public scores.

Table IV retains every rule. Top-ten intervals recover 27 censored effects; rank/pair-gap proxies falsely support 107/504 and 25/504 score nulls. Strict-null gating loses 29/72 detections and adds 116 insufficiencies without reducing errors. Proxy errors persist at top fifty despite visibility of every null target. Exact-zero null differences and minimum effect 0.00811687 make zero/guarded

point rules coincide. Themes, queries and orders are dependent.

APPENDIX C

NEW-QUERY DEPTH AND PRECISION

Eight equipment themes provide sixteen queries disjoint from attribution/development queries. Six edits are unchanged, case-only, date, scheduling, frequency and adverb. Two field policies and phases use $N = 128/512/1,024$, $k = 5/10/50$. Eighty topic-related records stay fixed as RFC backgrounds grow; 440/944 largest-corpus backgrounds overlap attribution. Encoder and components are reused.

Twelve worlds require 6,656 writes, 208 ingestion requests and 576 searches. Reconstruction matches all 12,480 public scores. The 1,728 paired episodes comprise 576 effects and 1,152 score nulls. Prospectively specified HALF_EVEN projections at 6/4/3/2 decimals use exact rational half-step intervals and the same native-score oracle. Five precision views reuse these episodes without service calls; they are not 8,640 independent observations.

At top five, intervals add fifteen supports on ten queries and raise determinate coverage from 111/192 to 126/192 for every N ; top ten/fifty resolve all episodes (Table V). RFC growth leaves counts unchanged. All 576 top-five cells have zero minimum target-to-cutoff distance and verified fifth/sixth-score ties; none has distance in $(\varepsilon, 10^{-4}]$. For a phase-zero omission, support requires $r_1 - c_0^{\text{rep}} > \varepsilon + e_1 + e_0$, using the other phase's visible score. Own-phase distance is insufficient.

The smallest natural effect is 1.940×10^{-5} , leaving near-guard effects untested. All 144 case-only edits change selected text without changing recorded vectors/scores: score equality does not establish nonprocessing. Four-decimal rounding eliminates all 999 no-effect conclusions; point treatment of 4/3/2-decimal reports creates 18/45/354 false no-effect verdicts relative to the unrounded estimand (Table VI).

Table VII applies the main article's precision rule (4). Exactness concerns the native-score estimand; binary16/binary32 alone bound neither normalization, accumulation nor input errors. Unknown API contracts

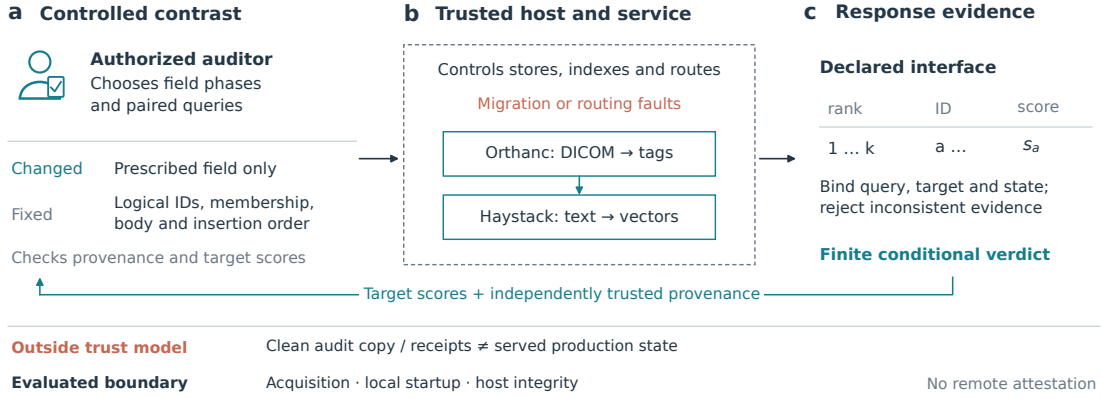


Fig. 5. Audit roles and evidence flow. (a) The authorized auditor fixes the contrast and paired queries. (b) The dashed boundary encloses the inspected host's stores and routing, which may differ from the intended configuration. (c) Within the evaluated trusted host, scores and provenance checks support a conditional verdict. The separate hostile-host example lies outside this guarantee.

TABLE III
ROLES AND TRUST IN THE EVALUATED HOST. CONFIGURATION, WRITE DESTINATIONS, AND READER BINDINGS ARE CHECKED UNDER TRUSTED ACQUISITION AND STARTUP; SELF-REPORTED LABELS DO NOT AUTHENTICATE A HOSTILE HOST.

Role	Permitted capability	Evidence boundary
Submitting outsider	Submit its own chosen input and ordinary queries; recognize its document only if the interface permits	No assumed control of corpus, reference documents, configuration, or traces
Authorized auditor	Specify field phases, paired targets, query set, contract, and decision rule; verify acquisition	Score verdict uses declared receipts and responses; unsupported contracts yield insufficient evidence
System operator	Control stores, readers, writes, and supplied receipts; realize the requested paired audit	Faulty configuration or routing is in scope; fabrication by the evidence-producing host is outside the guarantee

TABLE IV
SHARED-RESPONSE ATTRIBUTION AUDIT. AT EACH K, T IS SUPPORTED EFFECTS / 72, F FALSE SUPPORTS / 504 SCORE NULLS, AND I INSUFFICIENT / 576 EPISODES. OTHER CASES HAVE NO DETECTABLE EFFECT. THE ESTIMAND IS $|\Delta s_a| > 10^{-6}$; REPEATED QUERIES/ORDERS ARE DEPENDENT.

Method	T10	F10	I10	T50	F50	I50
Rank change	72	107	159	72	126	0
Pair-gap change	23	25	433	72	72	144
Target, nonzero	45	0	212	72	0	0
Target, guarded point	45	0	212	72	0	0
Target, interval	72	0	185	72	0	0
Interval + null gate	43	0	301	72	0	0

cannot be validated by saved-response projections. Depth cannot restore precision, and increasing ε changes the obligation.

a) *Engineered near-threshold stress*: At $N = 512$, one saved float32 target/query pair per query is interpolated toward 0, ± 0.25 , ± 0.5 , ± 1 , ± 2 , ± 4 times ε . Actual stock-cosine scores define labels; desired and achieved changes are all retained. The 192 calls comprise sixteen baselines and 176 constructions, without encoding: 87

effects, 73 nonzero within-tolerance changes and sixteen zeros. Quantization moves 23/32 intended boundary cases above the guard. Across three depths, 528 projections yield 261 effects and 267 nulls. Full-precision T/N/I is 218/221/89; six decimals give 80/139/309, with 55 false no-effect verdicts under point treatment. Four or fewer decimals leave all 528 insufficient. These constructions estimate neither natural prevalence nor metadata-ingestion behavior.

Identical selected texts can acquire unequal vectors under native batching: null score gaps reach 7.138×10^{-8} ; three IDs-only null positives arise from 4.167×10^{-9} . Such rank sensitivity is not a universal rule or numerical bound.

b) *Post hoc tolerance sensitivity*:

Saved intervals are reclassified at $\varepsilon \in \{0, 10^{-8}, 10^{-7}, 10^{-6}, 10^{-5}, 10^{-4}\}$, fixed before reanalysis but after collection, without new model/service calls or discarded cases. Each tolerance has its own oracle; more no-effect outcomes at a larger tolerance do not strengthen exclusion. At 10^{-6} , full precision leaves 45/576 natural and 43/261 engineered effects insufficient; six decimals leave

a Controlled phases				b Score bounds	c Whole-interval verdict
	Phase 0	Phase 1		Same target a; phases $j = 0, 1$	Guard $[-\varepsilon, +\varepsilon]$ (schematic)
	a	b	a	Visible, unrounded score	
Field swap	P	N	N	$I_j = [s_j, s_j]$	Supported effect wholly outside guard
Equal null	N	N	N	Absent from exact top-k	No detectable effect wholly inside guard
Reference only	N	P	N	$I_j = (-\infty, c_j]$	
Body mismatch*	N	N	N	c_j : certified cutoff	Insufficient evidence overlap or invalid evidence
P: topic-bearing; N: neutral				$I_\Delta = [l_1 - u_0, u_1 - l_0]$	
*Bodies differ between a and b only.				$\Delta = s_1(a) - s_0(a)$	
Fixed across phases				IDs · URLs · bodies · membership · query · insertion order	
Private validation				Field selection → selected text → complete score	Public path unresolved

Fig. 6. Paired score audit. (a) Four attribution cohorts vary topic-bearing (P) and neutral (N) comments while fixing each body across phases. (b) Unrounded visible scores and exact-search cutoffs bound the same target’s score change. (c) The whole difference interval determines the verdict relative to the guard; positions are schematic. Private validation and path diagnosis require separate evidence.

TABLE V

NEW-QUERY DEPTH STUDY, FULL PRECISION. EACH ROW HAS 192 EPISODES (64 EFFECTS, 128 SCORE-NULL COMPARISONS). T COUNTS TRUE SUPPORTED EFFECTS; I COUNTS INSUFFICIENT VERDICTS. BOTH METHODS HAVE ZERO FALSE SUPPORTS AND ZERO FALSE NO-EFFECT VERDICTS. NESTED CORPORA AND DEPTHS REUSE QUERIES; IDENTICAL ROWS ACROSS N ARE NOT INDEPENDENT REPLICATIONS.

N	k	Point T	Interval T	Point I	Interval I
128	5	34	49	81	66
128	10	64	64	0	0
128	50	64	64	0	0
512	5	34	49	81	66
512	10	64	64	0	0
512	50	64	64	0	0
1024	5	34	49	81	66
1024	10	64	64	0	0
1024	50	64	64	0	0

181/261 engineered effects insufficient. All decisive labels agree with the corresponding oracle, conditional on valid intervals.

APPENDIX D

FROZEN PUBLIC QUERIES AND EXACT SCORERS

SciFact [43] selection fixes the first sixteen numeric BEIR test-query IDs with positive qrels and their first two positive documents: eighteen pairs, fifteen targets. Nested corpora retain targets and earliest remaining IDs ($N = 128/512/5,183$), without score selection. Only target Titles change from *Administrative record*. to originals; bodies/membership stay fixed. Qrels indicate cited documents, not necessarily supporting evidence; topic/pretraining disjointness is unestablished.

Prefixes of 160 body and 48 title WordPiece tokens prevent MiniLM truncation. Title inclusion/exclusion,

depths 5/10/20/50 and $\varepsilon = 10^{-6}$ feed Haystack cosine and separately normalized float32 FAISS FlatIP [34], each with its own native complete-score oracle. Reconstruction checks 1,536 public responses, 384 complete responses and 864 paired cells; maximum discrepancies are 8.88×10^{-16} and 1.77×10^{-7} .

Both scorers increase determinate coverage from 385/432 to 389/432: four extra supports on four pairs, with no gain in nine of twelve conditions (Table VIII). Shared inputs do not give eight independent gains. Equal pair/policy/load/depth weighting gives two-target queries twice the weight of single-target queries.

APPENDIX E

EQUAL-ACCESS AND NATIVE MEASUREMENT PROTOCOLS

a) *Matched baseline*: Four unchanged 192-document stores cover two field worlds and

TABLE VI

KNOWN NEAREST-ROUNDING PROJECTIONS OF THE SAME 1,728 NEW-QUERY EPISODES (576 EFFECTS AND 1,152 SCORE NULLS). INTERVAL T, N, AND I ARE SUPPORTED EFFECTS, NO DETECTABLE EFFECT, AND INSUFFICIENT EVIDENCE. POINT FN COUNTS TRUE FULL-SCORE EFFECTS INCORRECTLY CALLED NO DETECTABLE EFFECT WHEN ROUNDED VISIBLE SCORES ARE TREATED AS EXACT. ALL INTERVAL ERRORS ARE ZERO BY VALID INTERVAL CONTAINMENT; THE MEASURED TRADEOFF IS ABSTENTION.

Decimal places	Interval T	Interval N	Interval I	Point FN
full	531	999	198	0
6	531	999	198	0
4	501	0	1227	18
3	336	0	1392	45
2	18	0	1710	354

TABLE VII

PRECISION PREFLIGHT FOR TWO VISIBLE SCORES, WITHOUT RANGE CLIPPING. WITH $d = r_1 - r_0$ AND $w = e_0 + e_1$, NO-EFFECT ACCEPTANCE REQUIRES $|d| + w \leq \varepsilon$; EFFECT SUPPORT REQUIRES $|d| > \varepsilon + w$. THE CONDITION $\varepsilon \geq w$ PERMITS, BUT DOES NOT GUARANTEE, ACCEPTANCE. PRECISION LIMITS ARE NOT REPAIRED BY INCREASING RETRIEVAL DEPTH.

Verified score contract	w	Acceptance feasible
Exact native scores	0	$\varepsilon \geq 0$
Nearest rounding, 8 decimals	10^{-8}	$\varepsilon \geq 10^{-8}$
Nearest rounding, 6 decimals	10^{-6}	$\varepsilon \geq 10^{-6}$
Nearest rounding, 4 decimals	10^{-4}	$\varepsilon \geq 10^{-4}$
Same rounding step h , additional errors η_0, η_1	$h + \eta_0 + \eta_1$	$\varepsilon \geq w$

two generations. Both methods inspect actual readers/descriptors and intended generation, then use the same target-score route for 24 frozen queries addressing two targets each. Direct comparison and singleton intervals apply the identical 10^{-6} rule.

The protocol and 29 dependency hashes were frozen before collection; earlier scientific outcomes were known. Each state has one warmup batch/method and six measured rounds, alternating AB/BA three times: four warmup and 24 measured batches. Each has 48 calls, 96 scores and 15,336 request-body bytes. Responses include binding envelopes but exclude headers. Shared startup takes 7.879 s; parsing, preparation, writing outputs, history acquisition and human effort are outside batch timing. The nonexclusive Xeon Platinum 8368Q host uses Python 3.12.3, Haystack 2.31.0, NumPy 2.5.3 and one native-scoring thread.

Direct/interval ranges overlap: 5.285–5.717/5.378–5.723 s before correction and 5.354–5.598/5.308–5.550 s afterward. Descriptor hashing precludes production-speed or scaling claims. Repetitions share 96 obligations; independent reconstruction checks binding, order and cosine for all 1,344 requests.

b) Upstream regression: Official LlamaIndex 0.12.27/0.12.28 wheels, sources and identical dependency locks are retained. The two-document fixture, query and collector were frozen before encoding

four strings with pinned MiniLM. An exact-string adapter supplies saved vectors while leaving native hashes and upsert decisions intact; it has its own transformation-cache identity and is not the stock HuggingFace integration. The old update retains metadata and vectors; the fixed release processes the edit. Upgrading alone, HTTP serving and generated answers were not tested. Sixteen native cosine outputs and saved states are independently checked.

c) Approximate-search protocol: HNSW uses normalized saved vectors, $M = 16$, efConstruction=100, seed 20260918, numeric-ID insertion and one thread. Four graphs serve sixteen queries for efSearch=8/32/128 and $k = 5/10/20/50$; each combination retains 64 responses and 36 cells. Privileged native scans enumerate all 5,183 scores/query to obtain the largest omitted score. Every condition remains, including zero-violation ef=32/ k =5 and ef=128/ k =5. Twelve naive target-containment failures repeat parameter cells (ten at ef=8, two at ef=32); observed decisive labels still agree with native truth. Direct target scores settle all 216 effects and 216 nulls. Saved graphs, neighbors, complete scores and timings support replay, not an unseen-state certificate.

APPENDIX F

CROSS-QUERY GEOMETRY: A NEGATIVE ADDITIONAL STUDY

a) Frozen design and equal information: After the first sixteen SciFact queries, we freeze all 284 remaining positive queries, their first two positive documents and all 5,183 documents: 305 pairs and 254 targets, with no target overlap with the previous fifteen. Both included phases freshly encode target inputs with administrative versus actual Titles under identical CPU, one-thread, batch-32 settings; unchanged documents and both excluded phases reuse original vectors. All 792 fresh inputs fit MiniLM (maximum 209 tokens). This joint intervention is a remaining subset of the same corpus after a negative pilot, not an independent dataset/deployment or external preregistration.

TABLE VIII

FROZEN SciFACT WORKLOAD. ENTRIES ARE DETERMINATE POINT / INTERVAL VERDICTS OUT OF 36 COMPARISONS (EIGHTEEN PAIRS, TWO FIELD POLICIES) AT EACH DEPTH. COUNTS COINCIDE FOR THE TWO SEPARATELY EVALUATED EXACT SCORERS. EXTRA T IS ADDITIONAL EFFECT SUPPORT SUMMED ACROSS THE FOUR DEPTHS PER ENGINE; ALL DECISIVE VERDICTS AGREE WITH THAT ENGINE'S COMPLETE-SCORE ORACLE.

Corpus size	k=5	k=10	k=20	k=50	Extra T
128	34 / 34	34 / 35	36 / 36	36 / 36	1
512	32 / 32	34 / 34	34 / 34	36 / 36	0
5,183	23 / 24	26 / 26	26 / 28	34 / 34	3

FlatIP and HNSW ($M = 16$, efConstruction=100, efSearch=32, seed 20260919) serve four policy/phase states at $k = 5/10/20/50$. Before private evaluation, A/B/P/C receive the same 284 query coordinates and 9,088 responses. Anchors are other queries visibly returning the target in the same engine/policy/phase/depth; private scores cannot select them. Query geometry and a trusted scorer contract exceed ordinary score-only access; separate public/private files do not enforce service permissions.

b) Known bounds and numerical contract: Let x be the stored target vector, q the objective query and A the visible-anchor query matrix. Coordinates are post-FAISS float32 values treated as exact binary rationals. Before new encoding/scoring we freeze binary64 $R = 1.0001$, $\eta = 10^{-4}$ and exact $\varepsilon = 1/1,000,000$. The actual vector must satisfy $\|x\|_2 \leq R$ and $l \leq Ax \leq u$; query norms obey R . The trusted contract bounds each native score's deviation from the real-coordinate dot product by η . Returned anchor score c_i gives an outward enclosure of $c_i \pm \eta$. For any finite λ , set $h = q - A^\top \lambda$:

$$\begin{aligned} L(\lambda) &= \sum_i \min\{\lambda_i l_i, \lambda_i u_i\} - R\|h\|_2, \\ U(\lambda) &= \sum_i \max\{\lambda_i l_i, \lambda_i u_i\} + R\|h\|_2. \end{aligned} \quad (5)$$

Expanding $q^\top x = \lambda^\top Ax + h^\top x$ and applying Cauchy–Schwarz proves (5): $L \leq q^\top x \leq U$, standard dual bounds without optimizer optimality. Missing native scores expand these real-dot bounds by η again before intersection with existing bounds. Visible native target scores stay singletons; the global range is $[-(R^2 + \eta), R^2 + \eta]$. Only exact-search omissions use cutoffs.

For $d = 384$, $u_{fp} = 2^{-24}$ and $\gamma_n = nu_{fp}/(1 - nu_{fp})$, the conventional allowance $\gamma_{2d+32}R^2 \approx 4.77 \times 10^{-5} < \eta$. Compiled normalization/dot-kernel conformance remains trusted; hashes and finite-run float64 comparisons do not verify every SIMD dispatch or define a universal error bound. Contract violations fail collection, not trigger post-hoc widening.

c) Candidates and independent verification: A uses per-query bounds. B adds known single-pivot candidates [35]; P constructs a nominal-center projection

baseline from classical multi-pivot geometry [36]. P's interval centers only propose coefficients; verification honors full uncertainty. C retains A/B/P and optimizes the interval support sum plus $R\|q - A^\top \lambda\|$ for q and $-q$ with zero/projection starts, bounded L-BFGS-B and at most 200 iterations. Neither optimality nor strict improvement is asserted.

A standard-library verifier computes support sums/squared residuals exactly, encloses square roots using integer square root with upward correction, then rounds binary64 endpoints outward. It caps coefficients at 10^8 and rejects malformed values or detectable norm/duplicate-anchor contradictions; these checks do not establish general joint feasibility. Invalid proposals retain baseline bounds; failure cells stay counted and cannot earn gains. Hashes bind packet identities/state, not authenticity. Separate phase intervals feed the main decision rule.

d) All results and negative diagnostics: A/B/P/C all give 1,957 effects, 1,943 no-effect verdicts and 980 insufficiencies across 4,880 dependent cells (Table IX). Of 1,957 omitted phase scores, 998 have anchors and narrower bounds; none changes a decision. Private native target scores resolve every cell. Independent checking of 5,988 certificates observes no containment/decisive-label errors or failed cells; this is a fixed-grid result, not independent trials or a statistical soundness estimate. Earlier 864-cell exact and 432-cell HNSW pilots also added zero decisions under provisional envelopes; they informed the design and are not prospective samples.

Post-hoc diagnostics find C narrows 542 phase intervals on 49 pairs beyond P, with median 2.91×10^{-8} , maximum 1.05×10^{-5} and 59 reductions above 10^{-6} . The larger median C-over-B reduction, 0.148422, is nearly entirely supplied by P. All 980 unresolved difference intervals contain zero, with minimum width 0.321882; private truth comprises 483 effects and 497 nulls. No operating point was changed. Over all 1,957 omitted phase scores, the median C-over-P width reduction is zero. The 980 unresolved difference intervals have median width 2.91579 and quartiles 2.26672/3.65033. These post-hoc quantiles use linear interpolation at sorted position $(n - 1)p$ and describe the fixed grid; repeated

TABLE IX

PROSPECTIVE CROSS-QUERY STUDY: ALL EIGHT FROZEN STRATA, 610 CELLS EACH. A, B, P AND C HAVE IDENTICAL T/N/I COUNTS IN EVERY ROW. MISSING COUNTS OMITTED TARGET PHASE SCORES; WITH ANCHORS COUNTS THOSE VISIBLE UNDER ANOTHER QUERY IN THE SAME ENGINE/POLICY/PHASE/DEPTH. ALL ANCHORED INTERVALS NARROW UNDER B, P AND C, BUT NONE CHANGES A VERDICT.

Engine	k	A/B/P/C T/N/I	Missing	With anchors
FlatIP	5	227 / 217 / 166	349	102
FlatIP	10	246 / 239 / 125	261	126
FlatIP	20	259 / 252 / 99	209	137
FlatIP	50	271 / 267 / 72	148	127
HNSW	5	209 / 215 / 186	355	101
HNSW	10	234 / 239 / 137	263	124
HNSW	20	247 / 250 / 113	215	135
HNSW	50	264 / 264 / 82	157	146

engine/policy/depth cells are not independent samples.

Native conversion alone has width at least 2η before intersection; this is not a lower bound after intersection or a general impossibility result. Cross-phase error correlations require another contract. Known encoder inputs, full order/tie constraints or sufficiently informative query geometry could add information, including vector recovery. The implemented outer bounds make no completeness claim.

e) Costs and scope: The common packet contains 193,120 scores and occupies 21,920,591 JSON bytes including coordinates/identities, not HTTP traffic. Combined analysis takes 70.965 s, including 62.671 s of certificate work; independent public verification takes 47.212 s before private evaluation. These are combined implementation timings, not per-method or production benchmarks. The 2,272 private full scans (11,775,776 scores) are excluded from auditor inputs and precede searches, potentially warming caches. No acquisition saving or added decision is established; independently enforced restricted-service access remains untested.